

Policy Atlas: Evaluation

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What is this paper?

An evaluation plan for assessing Policy Atlas.

Summary

This document presents an evaluation framework for Policy Atlas. The primary objective of this evaluation is to assess the tool's performance, validate the core capabilities and build an understanding of the level of confidence and accuracy of the results it produces to verify that it meets the criteria necessary for adoption by the target users. The findings from this evaluation will provide actionable data to guide the next phase of Policy Atlas development, prioritise improvements, and prepare for the roll-out to initial users.

1. Testing Framework

The evaluation will be assessing the following dimensions of the tool, to gain a holistic understanding of its performance and capabilities:

- **Accuracy:** The degree to which the tool's outputs correspond to a verifiable ground truth. This involves measuring the fidelity of its search, screening, extraction, assessment, and synthesis functions against gold-standard manual evidence reviews.
- **Robustness:** The consistency and reliability of the tool's performance across different contexts and its resilience to potential failure modes. This includes assessing for systematic bias, evaluating output stability, and testing its capacity for critical analysis and novel discovery.
- **Efficiency:** The quantifiable improvement the tool offers over manual workflows. This is primarily measured by the reduction in time and human effort required to achieve a comparable evidence synthesis output.
- **Utility:** The practical value, usability, and trustworthiness of the tool for its intended end-users. This dimension assesses how well the tool fits into the real-world workflows of target users and addresses their core pain points.

Each of these dimensions will be evaluated individually, to gain a holistic understanding of the tool's current performance.

2: Accuracy and Fidelity

This dimension of the evaluation conducts a direct, component-wise comparison of the tool's fidelity against the designated gold-standard reviews: AHL's Obesity Blueprint and the AFS' Parenting Support at Scale. The objective is to establish a quantitative and qualitative baseline of the tool's fidelity in replicating the core tasks of a systematic evidence review.

2.1 Reference Retrieval

Objective: To measure how effectively the Search feature identifies the evidence base of a manual systematic review.

Methodology: The evaluation will task Policy Atlas with replicating the search for the "Universal Free School Meals in Primary School" intervention from the Nesta Obesity

Blueprint. This will be a multi-step process:

1. **Query Generation Comparison:** Policy Atlas will be given a natural language query such as: "What is the effect of free school meals on obesity and calorie consumption in children?" The tool will convert this into a boolean search query for OpenAlex. This generated query will be qualitatively compared against the search strategy detailed in the relevant [search protocol](#) Blueprint document. The comparison will assess the alignment of key concepts, use of boolean operators (AND/OR), and overall query structure.
2. **Reference List Comparison:** Policy Atlas will execute its search across OpenAlex and Overton to generate a reference list. The DOIs and titles from this list will be compared against the bibliography (e.g. [Free School Meals](#)) of included and excluded studies from a Blueprint review.
3. **Recall Curve Analysis:** A core part of this evaluation is to understand the relationship between search depth and comprehensiveness. The search will be executed multiple times with an increasing number of requested documents (e.g., 25, 50, 100, 200...). For each run, recall against the gold-standard reference list will be calculated and plotted. This analysis will determine the optimal number of documents to retrieve to achieve a high level of coverage (e.g., >85% recall) and assess whether seminal works are captured in smaller, more rapid searches.

Metrics:

- **Recall (Primary Metric):** This metric will identify the percentage of the bibliography of documents from the manual review that Policy Atlas retrieves. It is the most critical metric for this component, as high recall is necessary to ensure the evidence base is comprehensive.
- **Seminal Paper Coverage:** As a sub-component of the above, the coverage of the top 5 most-cited papers from the Blueprint reference list will be identified. This metric will be the percentage of these seminal papers retrieved by Policy Atlas, with a particular focus on their inclusion in smaller, more rapid searches (e.g., when retrieving only 25 or 50 documents).
- **Precision:** This metric measures the efficiency of the search by answering: "Of all the documents Policy Atlas retrieved, what percentage were actually relevant (i.e., included in the gold-standard's final list)?" While less critical than recall, high precision indicates a well-targeted search that reduces the burden of screening irrelevant material.
- **F1-Score:** As the harmonic mean of precision and recall, the F1-score provides

a single, balanced measure of the overall accuracy of the reference retrieval process.

- **Query Structure Similarity:** A qualitative score (1-5) assessing the alignment between the LLM-generated query and the expert-crafted query from the gold-standard protocol, based on conceptual coverage and logical structure.
- **Recall vs. Document Count Curve:** A graphical plot demonstrating how recall improves as the number of retrieved documents increases, used to identify the point of diminishing returns.

Possibilities for Iteration: Based on the results of this evaluation, several parameters of the search function can be iterated upon to improve performance. These include:

- The number of papers retrieved by default.
- The number of parallel searches conducted.
- The stopping criteria for the search process.
- The methodology used to formulate queries from natural language input.

2.2 Relevance Screening

Objective: To evaluate the accuracy of the LLM-based screening of document abstracts and summaries against the decisions made during the gold-standard manual review process. The accuracy of this stage is essential to producing a valid evidence base.

Methodology: This evaluation will compare Policy Atlas's screening decisions against the ground-truth records from a Blueprint manual review ([e.g.](#)). Policy Atlas will be fed the same initial, unscreened list of documents retrieved in Component 2.1. The tool will then perform the relevance check, assigning:

- [yes/no] label,
- confidence score
- textual Reasoning for decision.

These outputs will be compared against the ground-truth labels from the human-led review, with the above context in mind. An "LLM-as-a-judge" approach will also be used to evaluate the quality of the reasoning, providing a consistent assessment of the model's logical coherence.

We should, however, acknowledge a difference in objectives between the two processes. The Blueprint review aimed to identify a single best paper for each recommendation, leading to a very strict filtration process at the screening stage. In contrast, Policy Atlas is designed to build an evidence base consisting of many relevant papers. This difference means that a direct comparison of specificity may be misleading. Policy Atlas may correctly identify a larger set of relevant documents that were excluded by the blueprint's stricter criteria, resulting in a lower specificity score that does not accurately reflect poor performance. To account for this, the evaluation will supplement quantitative metrics with a qualitative review. An expert assessment of a sample of documents included by Policy Atlas but excluded by the blueprint will determine if they are genuinely irrelevant or just outside the narrow scope of finding a "single best paper."

Metrics:

- **Accuracy:** The overall percentage of documents (both relevant and irrelevant) that Policy Atlas correctly classifies, providing a general measure of performance.
- **Sensitivity (Recall):** Measures the model's ability to correctly identify all relevant papers, minimising false negatives. High sensitivity is crucial - failing to include a relevant paper at this stage means it is permanently lost from the review.
- **Specificity:** This measures the model's ability to correctly reject irrelevant papers. This metric will be interpreted in the context of the differing screening goals. A qualitative review will be used to assess whether papers included by Policy Atlas but excluded by the blueprint are genuinely irrelevant or simply outside the blueprint's narrower scope.
- **LLM-as-a-Judge Consistency Score:** A numeric score (1-5) from a judge LLM (high reasoning model such as GPT-5), assessing the logical coherence between the screening model's decision score and its provided reasoning. This will be validated against a smaller, human-annotated dataset to ensure the judge's reliability.

Possibilities for Iteration:

- The LLM used for the screening task.
- The prompts provided to the screening model.
- The confidence score threshold used for automatic inclusion or exclusion of documents.
- The amount of context (e.g., abstract only vs. abstract and introduction) provided to the model to make its decision.

2.3 Information Extraction

Objective: To measure the accuracy and completeness of the tool's automated extraction of structured data points (key issues, interventions, outcomes, effect sizes) from the full text of relevant documents.

Methodology: A sample of full-text articles from the Blueprint evidence base will be selected, representing a mix of study types and complexities. The abstract screening sheets contain tabs for extractions which contain similar data points to what Policy Atlas extracts, so can function as the ground-truth dataset of key information: interventions, outcomes, and quantitative data points like effect size and p-value. One omission is that key issues were not extracted in the blueprint methodology so this will have to be done manually at this stage. Policy Atlas will then process the same articles, and its extracted data will be systematically compared to the human-produced ground truth.

To enhance the scalability of this evaluation, an LLM-as-a-judge methodology will be used alongside manual checks. For qualitative extractions, such as the narrative descriptions of interventions and outcomes, a judge LLM will be tasked with a reference-based scoring task. It will compare the tool's extracted text against the corresponding section in the source document and score its faithfulness, accuracy, and completeness on a numeric scale.

Metrics:

- **Match Rate:** The percentage of extracted entities (e.g. the specific names of interventions or outcomes) that correctly and completely match the entities in the ground-truth dataset.
- **Accuracy of Mapped Pairs:** For the entities that are successfully extracted, this metric assesses the percentage of Intervention-Outcome pairings that are correctly mapped as they were in the original study. This verifies that the tool understands the relationships between concepts, not just the concepts themselves.
- **Quantitative Data Accuracy:** For numerical data such as effect sizes, confidence intervals, and p-values, this metric will be calculated as the percentage of correctly extracted values.
- **Table and Chart Extraction Test:** A binary success/fail metric for each non-textual element in the sample, indicating whether Policy Atlas was able to extract the relevant data points contained within it.

- **LLM-as-a-Judge Faithfulness Score:** A score (1-5) assigned by the judge LLM, measuring how accurately the extracted narrative text reflects the source material. The judge will be prompted to provide step-by-step reasoning for its score to ensure transparency.

Possibilities for Iteration:

- The LLM used for the extraction tasks.
- The specificity of the extraction schema.
- Adding custom extraction criteria, e.g. mode of delivery, study length
- The techniques used for extracting data from non-textual elements like tables and charts.
- The prompts used to generate narrative descriptions of interventions and outcomes.

2.4 Judgement and Assessment

Objective: To assess how well Policy Atlas's automated scores for Evidence Strength and Predicted Impact align with the judgments of manual reviews. This tests the tool's analytical capabilities.

Methodology: Policy Atlas employs a 1-5 star rating for evidence strength, and predicted impact. This will be compared to the intervention assessments that were manually completed for [PSaS](#). The PA method assesses factors like study design (e.g. RCT, quasi-experimental), sample size, use of validated measures, and mitigation of confounders. In contrast, the PSaS assessment uses the [Foundations'](#) evidence standards, a scale (2-4+) that indicates confidence in causal evidence based on detailed methodological criteria. For example, Level 3 (efficacy) requires a rigorous evaluation demonstrating a statistically significant positive impact, while Level 4 (effectiveness) requires evidence from at least two such evaluations, including long-term outcomes. Given these different scales, a simple comparison is not sufficient. The evaluation will therefore use a two-part methodology:

1. **Quantitative Correlation:** A conceptual mapping will be established between the two scales (e.g., Foundations Level 4 $\approx$ 5 stars, Level 3 $\approx$ 4 stars). Based on this mapping, the rank ordering of interventions produced by Policy Atlas will be compared to the final ranking in the PSaS to test for statistical alignment.
2. **Qualitative Justification Analysis:** An expert will compare the textual justification provided by Policy Atlas for its star rating against the specific *criteria* met by the study to achieve its Foundations Level rating. This will verify

that the tool's reasoning is sound and conceptually aligned with expert judgment. For example, the analysis will check if a 5-star rating from Policy Atlas, justified by citing a large-sample RCT, corresponds to a study that meets the Foundations Level 3 or 4 criteria for rigorous experimental design.

Metrics:

- **Rank Correlation (Spearman's ρ):** A statistical measure (ρ) will be calculated to determine the correlation between the rank order of interventions as assessed by Policy Atlas and the rank order presented in the gold-standard review, based on the established mapping between the two rating scales. A high positive correlation would indicate strong alignment in overall judgment.
- **Top-Tier Agreement:** This metric calculates the percentage of agreement on the top 3 or top 5 highest-rated interventions. This focuses the assessment on the most policy-relevant findings, as these are the interventions most likely to be recommended.
- **Qualitative Justification Analysis:** An human reviewer will use a rubric to score the quality and accuracy of the LLM-generated justifications for its Evidence Strength and Predicted Impact ratings. This qualitative analysis will assess whether the tool's reasoning is sound, references the correct aspects of the study's methodology (e.g. sample size, randomisation), and aligns with the logic of human experts. This will also test whether the model is sufficiently critical compared to human judgement.
- **LLM-as-a-Judge Assessment Score:** A score (1-5) assigned by the judge LLM, measuring the extent to which the calculated evidence/impact scores and their explanations are consistent with the scoring methodology and source material. The judge will be prompted to provide step-by-step reasoning for its score to ensure transparency.

Possibilities for Iteration:

- The LLM used for the assessment task.
- The content and weighting of the rubrics for 'Evidence Strength' and 'Predicted Impact'.
- The prompts used to generate the textual justifications for the scores.
- A conceptual mapping between Policy Atlas's rating scale and other external evidence frameworks.

2.5 Synthesis

Objective: To evaluate the quality, accuracy, coherence, and utility of the final synthesised outputs, such as the executive summary and clustered intervention lists. This component assesses the end-product that will be consumed by the target users.

Methodology: A reviewer will conduct a side-by-side comparison of the Policy Atlas-generated summary and the original human-authored summary for a Blueprint recommendation. We could additionally compare an LLM generated summary of the PSaS report. The Policy Atlas summary is designed as a concise, high-level overview, whereas the original gold-standard summaries are more in-depth narrative reports. The evaluation will therefore focus not on a direct replication of text, but on whether the Policy Atlas summary accurately captures the conclusions, key takeaways, and overall recommendations of the more detailed source material. Additionally, a fact-checking process will be undertaken for the Policy Atlas summary, where each recommendation is traced back to a source document in the retrieved evidence base to guard against LLM hallucination.

Metrics:

- **Narrative Alignment Score:** A qualitative score (1-5 scale) from the human reviewer assessing how well the overall conclusions, key takeaways, and narrative of the Policy Atlas summary match those of the original gold-standard review (Or high reasoning LLM generated summary), accounting for the difference in format and conciseness.
- **Factual Grounding Score:** A quantitative metric representing the percentage of factual claims made in the Policy Atlas executive summary that can be verifiably traced to a specific statement or data point in one of the source documents.
- **Utility Score:** A qualitative score (1-5) that assesses the generated summary from the perspective of a policy-maker, scoring will be based on factors such as:
 - Clarity
 - Actionability
 - Conciseness
 - Traceability of claims
- **Cluster Coherence Rating (1-5):** The score indicating the internal coherence of the thematic issue/intervention clusters.
- **Cluster Label Rating (1-5):** The score indicating the quality of the thematic cluster labels.

Possibilities for Iteration:

- The LLM used for generating the executive summary.
- The clustering workflow used to group similar interventions.
- The length and structure of the final summary to better suit different user personas.
- The prompts that guide the narrative focus of the summary.

3. Robustness and Reliability

This aspect of the evaluation is designed to assess the system's stability and consistency, investigate for systematic biases, and uncover potential failure modes.

3.1 Stability and Consistency

Objective: To analyse the impact of model stochasticity on the reliability and consistency of the results. For the tool to be trustworthy, users must have confidence that the outputs are stable.

Methodology: A single, complex research query will be run through the Policy Atlas five separate times. The outputs at each key stage of the process will be collected for all five runs and compared to measure the degree of variation. The test will then be repeated, this time introducing slight variations in the query for each search to test how much small differences in the input affect the variability of the output.

Metrics:

- **Jaccard Index for Reference Lists:** This metric will be used to measure the overlap of the retrieved document sets between runs. A high Jaccard index would indicate a stable and deterministic search and screening process.
- **Variance in Quantitative Extractions:** For numerical data points (e.g., effect sizes) extracted from a common set of papers across the runs, the number of successfully extracted data points will be tracked and compared across runs for any variation.
- **Semantic Similarity for Summaries:** The final executive summaries generated in each of the five runs will be compared using semantic similarity metrics (e.g., BLEU). This will provide a quantitative measure of the consistency of the final narrative synthesis, assessing whether the key messages and conclusions remain stable across runs.

3.2 Representativeness

Objective: To investigate whether the tool's end-to-end workflow introduces biases.

Methodology: The analysis will examine the distribution of sources in the collected evidence base across several dimensions:

- **Geography:** The country or region of the study's origin.
- **Language:** The original language of publication.
- **Methodology:** The type of study (e.g. RCT, quasi-experimental, qualitative study, theoretical paper).
- **Publication Type:** The type of source (e.g. academic journal, government report, think tank publication).
- **Publication Date:** The year of publication, to assess for temporal bias.

The distribution of sources in the Policy Atlas output will be compared against two baselines: 1) the distribution within an original gold-standard review, and 2) the baseline distribution of documents, pre-filtering within the source databases ([OpenAlex](#) and [Overton](#)) for a given topic. This allows for the differentiation between bias inherent in the source literature and bias introduced by Policy Atlas.

In addition to this quantitative analysis, a qualitative review will be conducted to identify more subtle forms of bias. The review will be assessing the following:

- **Thematic Bias:** Does the tool systematically favor certain theoretical frameworks, intervention types (e.g. demand-side vs. supply-side), or ideological perspectives present in the literature?
- **Framing and Tone:** Does the final synthesis adopt a particular framing (e.g. overly positive or critical) that is not representative of the balance of findings in the underlying evidence?
- **Recency Bias:** Does the synthesis give more weight to recent publications at the expense of foundational or seminal older studies?
- **Omission of Nuance:** Does the tool oversimplify or omit important nuances, caveats, or conflicting findings reported in the source documents?

3.3 Assessment of Additional Evidence

Objective: To assess additional evidence that has been identified as relevant by Policy Atlas that was not included in the gold-standard manual review. This analysis serves to evaluate the nature and quality of these additional findings.

Methodology: A list will be compiled of all relevant documents that were retrieved and screened-in by Policy Atlas but were *not* included in the final bibliography of the gold-standard review. This list of additional documents will then be reviewed by a subject matter expert for a secondary review. The expert will assess each additional document for its relevance to the research question and its methodological quality, using a predefined rubric.

Metrics:

- **Additional Evidence Rate (%):** The percentage of the total corpus of relevant documents identified by Policy Atlas that were not present in the original gold-standard review. This provides a measure of the volume of additional information surfaced beyond the baseline review.
- **High-Quality Additional Evidence Rate:** Of the documents identified as additional, this metric calculates the percentage that an expert reviewer deems to be both highly relevant and methodologically sound.

3.4 Critical Assessment

A crucial aspect of robustness is the tool's capacity for critical assessment. LLMs can tend towards being sycophantic, uncritically summarising a paper's claims without evaluating their validity. The Policy Atlas must be able to differentiate between well-supported claims and those from methodologically flawed studies. To test this, the evaluation must go beyond the high-quality papers found in the gold-standard reviews. A challenge set of documents will be curated, containing studies with known methodological weaknesses (e.g., small sample sizes, p-hacking, conclusions that overstate the evidence). This challenge set will be processed by Policy Atlas, and the evaluation will specifically measure whether the "Evidence Strength" assessment correctly identifies these flaws, assigns a low score, and provides an accurate justification for doing so.

4. Efficiency and Utility

4.1 Time-to-Insight

Objective: To provide a measure of the efficiency gains offered by Policy Atlas when compared to manual evidence review methodologies.

Methodology: A baseline for the time and resources required for a manual review will be established using project management records from the gold-standard

projects. For example, the internal AFS Parenting Interventions Review provides a concrete data point, having required 106 days of work from a team of analysts and researchers. This baseline in person-hours will be compared against the end-to-end time required for Policy Atlas to produce a comparable output for the same research question.

Metrics:

- **Total Time-to-Output:** The time from the initial submission of a query to the generation of the final, synthesised output by Policy Atlas.
- **Output Match Estimate:** A qualitative estimate of the percentage match in synthesis quality when compared to the manual gold-standard review. Gives an indication of the amount of additional manual work that would be required post Policy Atlas to produce a comparable output.
- **Estimated Person-Hour Savings:** A calculation of the percentage reduction in human effort. This will be calculated as $(\text{Manual Person-Hours} - \text{Policy Atlas Person-Hours}) / \text{Manual Person-Hours} / \text{Output Match Estimate}$.

4.2 Cost of insight

How much does a complex query cost?

4.3 Comparative benchmarking

Objective: To conduct a direct comparison of Policy Atlas against similar tools on a standardised evidence synthesis task. This will provide objective data on its relative strengths and weaknesses.

Methodology: A single, well-defined research question (e.g., "What are the most effective interventions to improve early years literacy outcomes?") will be selected. This question will be run through Policy Atlas and its primary competitors. The competitors represent different segments of the market:

1. **Specialised Systematic Review Platforms:** Elicit, Otto-SR, Consensus, Undermind
2. **Generalist "Deep Research" Tools:** A state-of-the-art model from providers like OpenAI, Google (Gemini)

The outputs from each platform will be systematically reviewed and scored. The comparison will be structured using a feature and performance matrix.

Metrics:

- **Feature Completeness Score:** A quantitative score will be assigned to each tool based on the presence, absence, and quality of key features deemed useful for the policymaking use case. This includes features like integrated grey literature sources (Overton), explicit evidence and impact scoring, a structured summary format, and user controls for the search and synthesis process.
- **Output Quality Score (Blind Review):** Reviewers will assign a score (on a 1-5 scale) to the final synthesised output from each tool, without knowing which tool it came from. This score will be a composite measure of accuracy (verifiability of claims), comprehensiveness (coverage of key evidence), and actionability (clarity and utility for a policy audience).

4.3 User Workflow

Objective: To gather qualitative data on the tool's usability, trustworthiness, and overall fit within the workflows of its primary users

Metrics:

- **Task Success Rate (%):** The percentage of users who are able to successfully complete their assigned task within a reasonable timeframe using the tool.
- **Qualitative Feedback Themes:** The interview transcripts and think-aloud recordings will be coded to identify recurring themes. These themes will be analyzed to understand user perceptions regarding:
 - **Trust and Reliability:** Do users trust the outputs? What features enhance or detract from their confidence?
 - **Clarity and Actionability:** Do users understand the synthesised outputs? Can they quickly identify the "so what" for policy?
 - **Workflow Integration:** How would this tool change their current process? Does it solve their identified pain points, such as difficulty navigating fragmented evidence or the time-consuming nature of manual synthesis?

- **Feature Utility:** Which features (e.g. evidence strength scores, intervention clustering) do users find most and least valuable?